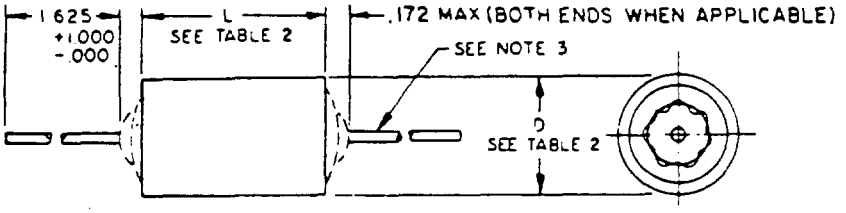
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, TUBULAR (INSULATED) (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CQR09

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-19978.



INCHES	MM	INCHES	MM
.001	.03	.312	7.92
.004	.10	.400	10.16
.025	.64	.562	14.27
.032	.81	1.000	25.40
.172	4.37	1.562	39.67
.235	5.97	1.625	41.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned solid wire, .025(No. 22 AWG) for cases .235 and .312 in diameter; and .032(No. 20 AWG) for cases .400 diameter and above. Tolerance on all lead wire diameters shall be +.004, -.001.
4. Capacitors with dimension L of 1.562 or D of .562 and larger, are not intended to be supported by their leads. These capacitors shall be provided with a supplementary means of mounting, such as a wrap-around band.

FIGURE 1. Style CQR09 capacitors.

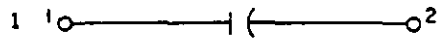
FSC 5910

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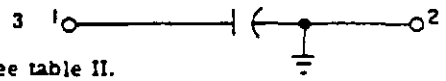
REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and table II.
Case material - Nonmagnetic (end seal may be of magnetic material).
Dielectric material - Paper and polyethelene terephthalate (K).
Terminals - Axial-wire lead (see figure 1).

Circuit diagram - 1



3



Rated voltage - See table II.
Rated temperature - -65° to +125°C.
Capacitance (Cap.) (nom) - See table II.
Capacitance tolerance - See table II.
Dissipation factor (DF) (max) - 1.0 percent.
Failure rate level - In accordance with MIL-C-19978.
Reliability - In accordance with MIL-C-19978.

Burn-in: In accordance with MIL-C-19978.

Radiographic inspection: In accordance with MIL-C-19978.

Seal: Method 112 of MIL-STD-202, test condition letter - A.

Dielectric withstanding voltage (DWV):
Sleeving - In accordance with MIL-C-19978.

Barometric pressure: In accordance with MIL-C-19978.
Test points:
Circuit diagram 1 - Between terminals and case.
Circuit diagram 3 - Between ungrounded terminal and case.

Insulation resistance (IR):
Sleeving - In accordance with MIL-C-19978.
Terminal to terminal - See table I.

TABLE I. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	
	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case - Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:
Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

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Salt spray: In accordance with MIL-C-19978.

Immersion:
DWV:
Sleeving - In accordance with MIL-C-19978.
IR:
Sleeving - In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter - D.

Moisture resistance:
DWV:
Sleeving - In accordance with MIL-C-19978.
IR:
Sleeving - In accordance with MIL-C-19978.

Stability at low and high temperature:
Low temperature:
Test temperature - $-65^{\circ}\ {}^{+0^{\circ}}_{-5^{\circ}}$C.

Capacitance change (max) - -10 percent.
High temperature:
Test temperature - $+125^{\circ}\ {}^{+5^{\circ}}_{-0^{\circ}}$C.

Capacitance change (max) - +10 percent.

Life:
Capacitance change (max) - ±5 percent of initial measured value.

Resistance to soldering heat: In accordance with MIL-C-19978.

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TABLE II. Style CQR09 capacitors.

Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate levels	Case dimensions 2/ Circuit 1 L ±.031	Circuit 1 D -.015 -.005	Circuit 3 L ±.031	Circuit 3 D -.015 -.005
	Volts	µF			Inches	Inches	Inches	Inches
CQR09A-KC392-3-	200	.0039	G, J, K	L,M,P,R,S	.750	.235	.688	.235
CQR09A-KC472-3-	200	.0047	G, J, K					
CQR09A-KC562-3-	200	.0056	G, J, K					
CQR09A-KC682-3-	200	.0068	G, J, K					
CQR09A-KC183-3-	200	.018	G, J, K		.875	.312	.812	.312
CQR09A-KC223-3-	200	.022	G, J, K					
CQR09A-KC273-3-	200	.027	G, J, K					
CQR09A-KC333-3-	200	.033	G, J, K					
CQR09A-KC393-3-	200	.039	G, J, K					
CQR09A-KC473-3-	200	.047	G, J, K		.875	.400	.812	.400
CQR09A-KC563-3-	200	.056	G, J, K					
CQR09A-KC683-3-	200	.068	G, J, K					
CQR09A-KC823-3-	200	.082	G, J, K					
CQR09A-KC104-3-	200	.10	G, J, K		1.125	.400	1.062	.400
CQR09A-KC124-3-	200	.12	G, J, K		1.375	.400	1.312	.400
CQR09A-KC154-3-	200	.15	G, J, K					
CQR09A-KC184-3-	200	.18	G, J, K		1.125	.562	1.062	.562
CQR09A-KC224-3-	200	.22	G, J, K					
CQR09A-KC274-3-	200	.27	G, J, K		1.375	.562	1.312	.562
CQR09A-KC334-3-	200	.33	G, J, K					
CQR09A-KC394-3-	200	.39	G, J, K					
CQR09A-KC474-3-	200	.47	G, J, K		1.625	.562	1.562	.562
CQR09A-KC564-3-	200	.56	G, J, K					
CQR09A-KC684-3-	200	.68	G, J, K		1.625	.670	1.562	.670
CQR09A-KC824-3-	200	.82	G, J, K					
CQR09A-KC105-3-	200	1.000	G, J, K		2.125	.750	2.062	.750
CQR09A-KE272-3-	400	.0027	G, J, K		.750	.235	.688	.235
CQR09A-KE332-3-	400	.0033	G, J, K					
CQR09A-KE123-3-	400	.012	G, J, K		.875	.312	.812	.312
CQR09A-KE153-3-	400	.015	G, J, K					
CQR09A-KE273-3-	400	.027	G, J, K		.875	.400	.812	.400
CQR09A-KE333-3-	400	.033	G, J, K					
CQR09A-KE393-3-	400	.039	G, J, K		1.125	.400	1.062	.400
CQR09A-KE473-3-	400	.047	G, J, K					
CQR09A-KE563-3-	400	.056	G, J, K		1.375	.400	1.312	.400
CQR09A-KE683-3-	400	.068	G, J, K					
CQR09A-KE823-3-	400	.082	G, J, K		1.125	.562	1.062	.562
CQR09A-KE104-3-	400	.10	G, J, K					
CQR09A-KE124-3-	400	.12	G, J, K		1.375	.562	1.312	.562
CQR09A-KE154-3-	400	.15	G, J, K					
CQR09A-KE184-3-	400	.18	G, J, K		1.625	.562	1.562	.562
CQR09A-KE224-3-	400	.22	G, J, K					
CQR09A-KE274-3-	400	.27	G, J, K		1.625	.670	1.562	.670
CQR09A-KE334-3-	400	.33	G, J, K					
CQR09A-KE394-3-	400	.39	G, J, K					
CQR09A-KE474-3-	400	.47	G, J, K	L,M,P,R,S	2.125	.750	2.062	.750

See footnotes at end of table.

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TABLE II. Style CQR09 capacitors. continued

Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate levels	Case dimensions 2/ Circuit 1 L±.031	Circuit 1 D+.015 -.005	Circuit 3 L±.031	Circuit 3 D+.015 -.005
	Volts	µF			Inches	Inches	Inches	Inches
CQR09A-KF102-3 -	600	.0010	G, J, K	L,M,P,R,S				
CQR09A-KF122-3 -	600	.0012	G, J, K					
CQR09A-KF152-3 -	600	.0015	G, J, K		.750	.235	.688	.235
CQR09A-KF182-3 -	600	.0018	G, J, K					
CQR09A-KF222-3 -	600	.0022	G, J, K					
CQR09A-KF272-3 -	600	.0027	G, J, K					
CQR09A-KF332-3 -	600	.0033	G, J, K					
CQR09A-KF392-3 -	600	.0039	G, J, K					
CQR09A-KF472-3 -	600	.0047	G, J, K		.875	.312	.812	.312
CQR09A-KF562-3 -	600	.0056	G, J, K					
CQR09A-KF682-3 -	600	.0068	G, J, K					
CQR09A-KF822-3 -	600	.0082	G, J, K					
CQR09A-KF103-3 -	600	.010	G, J, K					
CQR09A-KF123-3 -	600	.012	G. J. K					
CQR09A-KF153-3 -	600	.015	G, J, K		.875	.400	.812	.400
CQR09A-KF183-3 -	600	.018	G, J, K					
CQR09A-KF223-3 -	600	.022	G, J, K					
CQR09A-KF273-3 -	600	.027	G, J, K		1.125	.400	1.062	.400
CQR09A-KF333-3 -	600	.033	G, J, K					
CQR09A-KF393-3 -	600	.039	G, J, K		1.375	.400	1.312	.400
CQR09A-KF473-3 -	600	.047	G, J, K					
CQR09A-KF563-3 -	600	.056	G, J, K		1.125	.562	1.062	.562
CQR09A-KF683-3 -	600	.068	G, J, K					
CQR09A-KF823-3 -	600	.082	G, J, K		1.375	.562	1.312	.562
CQR09A-KF104-3 -	600	.10	G, J, K					
CQR09A-KF124-3 -	600	.12	G, J, K		1.625	.562	1.562	.562
CQR09A-KF154-3 -	600	.15	G, J, K					
CQR09A-KF184-3 -	600	.18	G, J, K		1.625	.670	1.562	.670
CQR09A-KF224-3 -	600	.22	G, J, K					
CQR09A-KF274-3 -	600	.27	G, J, K		2.125	.750	2.062	.750
CQR09A-KF334-3 -	600	.33	G, J, K					
CQR09A-KF394-3 -	600	.39	G, J, K		2.375	.750	2.312	.750
CQR09A-KF474-3 -	600	.47	G, J, K					
CQR09A-KG102-3 -	1000	.0010	G, J, K					
CQR09A-KG122-3 -	1000	.0012	G, J, K					
CQR09A-KG152-3 -	1000	.0015	G, J, K					
CQR09A-KG182-3 -	1000	.0018	G, J, K					
CQR09A-KG222-3 -	1000	.0022	G, J, K					
CQR09A-KG272-3 -	1000	.0027	G, J, K					
CQR09A-KG332-3 -	1000	.0033	G, J, K		.875	.400	.812	.400
CQR09A-KG392-3 -	1000	.0039	G, J, K					
CQR09A-KG472-3 -	1000	.0047	G, J, K					
CQR09A-KG562-3 -	1000	.0056	G, J, K					
CQR09A-KG682-3 -	1000	.0068	G, J, K					
CQR09A-KG822-3 -	1000	.0082	G, J, K					
CQR09A-KG103-3 -	1000	.010	G, J, K	L,M,P,R,S				

See footnotes at end of table.

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TABLE II. Style CQR09 capacitors. continued

Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate levels	Case dimensions 2/ Circuit 1 L±.031	Circuit 1 D+.015 -.005	Circuit 3 L±.031	Circuit 3 D+.015 -.005
	Volts	µF			Inches	Inches	Inches	Inches
CQR09A-KG123-3-	1000	.012	G, J, K	L, M, P, R, S	1.125	.400	1.062	.400
CQR09A-KG153-3-	1000	.015	G, J, K					
CQR09A-KG183-3-	1000	.018	G, J, K		1.375	.400	1.312	.400
CQR09A-KG223-3-	1000	.022	G, J, K					
CQR09A-KG273-3-	1000	.027	G, J, K		1.125	.562	1.062	.562
CQR09A-KG333-3-	1000	.033	G, J, K					
CQR09A-KG393-3-	1000	.039	G, J, K		1.375	.562	1.312	.562
CQR09A-KG473-3-	1000	.047	G, J, K					
CQR09A-KG563-3-	1000	.056	G, J, K		1.625	.562	1.562	.562
CQR09A-KG683-3-	1000	.068	G, J, K					
CQR09A-KG823-3-	1000	.082	G, J, K		1.625	.670	1.562	.670
CQR09A-KG104-3-	1000	.10	G, J, K					
CQR09A-KG124-3-	1000	.12	G, J, K		1.875	.670	1.812	.670
CQR09A-KG154-3-	1000	.15	G, J, K					
CQR09A-KG184-3-	1000	.18	G, J, K		2.125	.750	2.062	.750
CQR09A-KG224-3-	1000	.22	G, J, K					
CQR09A-KG274-3-	1000	.27	G, J, K		2.125	1.0	2.062	1.000
CQR09A-KG334-3-	1000	.33	G, J, K					
CQR09A-KG394-3-	1000	.39	G, J, K		2.375	1.0	2.312	1.000
CQR09A-KG474-3-	1000	.47	G, J, K	L, M, P, R, S				

1/ Complete part number shall include additional symbols to indicate circuit, capacitance tolerance, and failure rate level as applicable.

2/ Dimensions are for basic case; sleeving shall extend 0.016 inch minimum, 0.062 inch maximum, beyond each end of the capacitor body; however, if a shrink-fitted insulation is used for the sleeving, it shall lap over the ends of the capacitor body. Add 0.047 inch maximum to the nominal for capacitor diameter.

TABLE III. Millimeter equivalents of decimal inches.

Inches	mm	Inches	mm
0.005	0.13	0.812	20.62
0.015	0.38	0.875	22.23
0.016	0.41	1.000	25.40
0.031	0.79	1.062	26.97
0.047	1.19	1.125	28.58
0.062	1.57	1.312	33.32
0.235	5.97	1.375	34.93
0.312	7.92	1.562	39.67
0.400	10.16	1.625	41.28
0.562	14.27	2.062	52.37
0.670	17.02	2.125	53.98
0.688	17.48	2.312	58.72
0.750	19.05	2.375	60.33

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Navy - EC
Air Force - 85

Review activities:
Army - MI, MU
Navy - AS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC, AS, OS, SH
Air Force -

Preparing activity:
Air Force - 85

Agent:
DSA - ES

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(See Instructions – Reverse Side)

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	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. *Paragraph Number and Wording:*

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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